

Muhammad Raihan Molla

AI Engineer · ML Researcher · Mobile & Backend Developer

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PROFESSIONAL SUMMARY

Final-year Computer Science and Engineering (CSE) student at BUBT with a strong foundation in machine learning research, deep learning. Experience in building cross-platform mobile applications (Flutter/Dart) and backend systems (Python, Django, FastAPI). Analytical and self-driven with a passion for translating research into real-world AI products.

TECHNICAL SKILLS

AI / ML Frameworks: PyTorch, TensorFlow/Keras, Scikit-learn, Hugging Face Transformers, YOLO, OpenCV
Deep Learning: CNN, EfficientNet, ResNet, Vision Transformers (ViT), LSTM, BiLSTM, Transformer architectures
Generative AI & LLMs: Prompt engineering, LangChain, ChromaDB (Vector DB), RAG pipelines,
Computer Vision: Image classification, object detection, deepfake detection, medical image analysis, FFT-based forensics
NLP: Text classification, sequence modelling, plagiarism detection, multilingual NLP (Bengali/English)
MLOps & Tools: Docker, Google Colab, Weights & Biases, Git/GitHub, Linux (Kali, Ubuntu), VS Code, Cursor, Claude Code
Programming: Python (primary), Dart, Java, C, C++, OOP and DSA
Backend : Basic FastAPI, Firebase Realtime Database, MongoDB, SQL
Mobile Development: Flutter/Dart, Flame Engine, Riverpod, cross-platform Android/iOS
Embedded / Edge AI: ESP32-CAM, Hugging Face model deployment on edge hardware
Graphics & Simulation: OpenGL, GLSL shaders, particle systems, physically based rendering

EDUCATION

Bangladesh University of Business & Technology (BUBT) Jan 2023 – 2027
B.Sc. Engineering in Computer Science & Engineering | Dhaka, Bangladesh
Govt. Ainuddin College 2019 – 2021
Higher Secondary Certificate (HSC), Science | GPA: 3.75
Madhukhali Pilot High School 2014 – 2019
Secondary School Certificate (SSC), Science | GPA: 3.50

RESEARCH & PUBLICATIONS

Deepfake Image Detection — Cross-Dataset Generalization Study Thesis

- Developed a dual-stream EfficientNet-B4 model fusing spatial and FFT-frequency features for deepfake detection.
- Evaluated on FF++, Celeb-DF, and DFDC datasets; studied cross-dataset generalization and domain shift.
- Implemented GPU-accelerated training pipeline with data augmentation and adversarial robustness testing.

AQI Prediction — LSTM/BiLSTM vs. Classical ML Baselines

- Compared LSTM, BiLSTM, ARIMA, Random Forest, and XGBoost on hourly pollutant and meteorological data.
- Achieved best performance with BiLSTM for multi-step air quality index forecasting.

Lung Cancer Detection Using Deep Learning on CT Scans Ongoing

- Applied CNN-based classification to CT scan datasets for early-stage lung cancer detection.
- Benchmarked multiple architectures (ResNet, VGG, custom CNN) on public medical imaging datasets.

PROJECTS

AI-Powered Plagiarism Detection [GitHub ↗](#) | [Python / NLP](#)

- Built a document-level plagiarism detection system using semantic embeddings and cosine similarity.
- Supports multi-document batch comparison with a REST API interface.

Incursion — Real-Time Black Hole Lifecycle Simulator

Python / OpenGL / GLSL

- Built a real-time astrophysical simulation with 27 custom GLSL shaders and a 30,000-particle system.
- Simulated gravitational lensing, accretion discs, relativistic jets, and Hawking radiation; presented to academic panel.

Mobile Apps — ARI Software Lab

Flutter / Dart / Firebase

- Published PDF Scanner, Money Manager (expense tracker with Bengali localization), Music Player, and AnimalCam AR camera app on Google Play.

WORK EXPERIENCE

ARI Software Lab — Founder & Lead Developer

2025 – Present

- Founded an independent software publishing mobile apps and prototypes to Google Play.
- Managed full product lifecycle: ideation, development, ASO optimization, Play Store publishing, and post-launch iteration.
- Published 3 apps on Google Play with 1000+ downloads.

RESEARCH INTERESTS

AI Safety • Deepfake Detection • Mechanistic Interpretability • Adversarial Robustness • Edge AI • Computer Vision • LLM Agent Architectures • Big Data